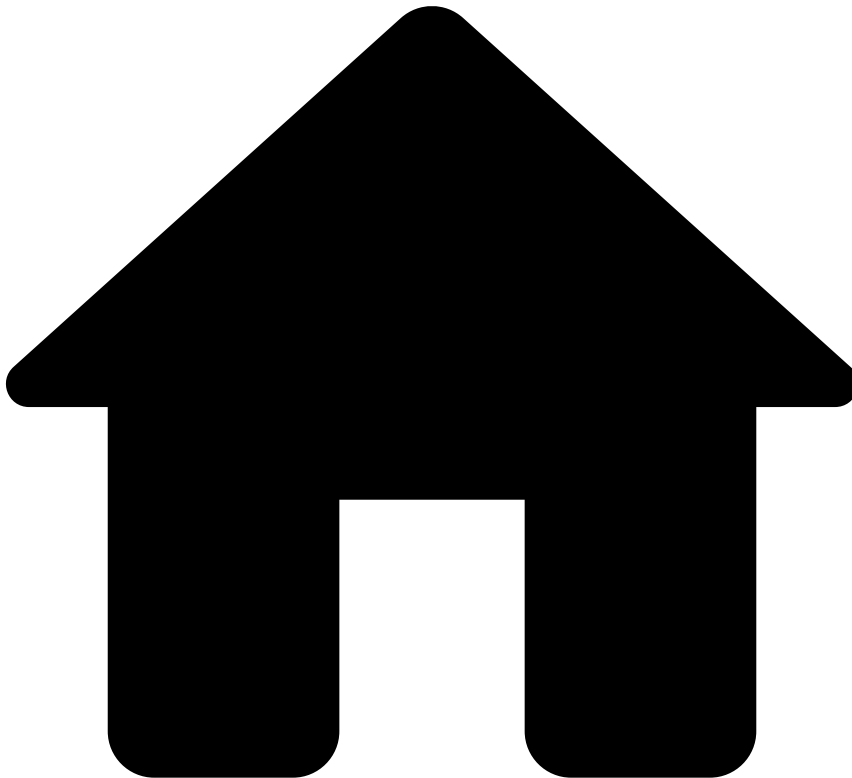


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# Transfers - Entity Limit Rules

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## Description

*Entity Limit* rules have the purpose of limiting the financial amount and number of transactions. The majority of these rules compare the recent user account behavior with past, long-term behaviors and define trigger conditions based on thresholds. These thresholds compare how much the transaction amount or number of transactions has increased compared to the past.

## Rules Project Metadata

- *Target Variable:* risk\_level

- *Tenancy Type:* SINGLE\_TENANT

## Rules in this rules project:

### Transfers-EL1

#### Comment

Daily amount per sender exceeds threshold

#### Actions

- [Transfers-EL1]

#### Variables

```
min_sum_amount_sender_24h_th
=transfers__rules_thresholds["min_sum_amount_sender_24h_th"].threshold;

formatted_sum = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sum_amount_per_sender_24h);

formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency,
min_sum_amount_sender_24h_th);
```

#### Condition

```
event.sum_amount_per_sender_24h >= min_sum_amount_sender_24h_th
```

#### Explanation

[Transfers-EL1] - Medium Risk - Sum amount ({localvar.formatted\_sum}) by this Sender in last 24h is above {localvar.formatted\_th}.

#### Mark As

The outcome of this rule is: **medium**

### Transfers-EL2

#### Comment

Sender's amount is significantly higher than daily average amount (considering last week's behavior)

#### Actions

- [Transfers-EL2]

#### Variables

```
min_ratio_sender_trx_amount_sender_daily_avg_th
=transfers__rules_thresholds["min_ratio_sender_trx_amount_sender_daily_avg_th"].threshold;

ratio=Maths.safeDivision((event.sender_transaction_amount ?? 0)*1.0,
event.avg_daily_amount_per_sender_1w);

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
```

## Condition

```
ratio >= min_ratio_sender_trx_amount_sender_daily_avg_th
```

## Explanation

[Transfers-EL2] - High Risk - Transaction amount ( $\{localvar.formatted\_amount\}$ ) is significantly higher ( $\{localvar.min\_ratio\_sender\_trx\_amount\_sender\_daily\_avg\_th\} \times$ ) than Sender's daily average amount (considering last week's behavior).

## Mark As

The outcome of this rule is: **high**

## Transfers-EL3

## Comment

Sender's monthly number of transactions is extremely higher than average number of transaction (considering behavior from previous 3 months)

## Actions

- [Transfers-EL3]

## Variables

```
min_ratio =
transfers__rules_thresholds["min_ratio_sender_trx_count_sender_monthly_avg_th"].threshold;
min_count = transfers__rules_thresholds["min_count_trx_sender_1m_th"].threshold;

ratio = match
  if (event.avg_monthly_count_sender_3m_de == 0.0) then
    return event.count_sender_1m;
  else
    return
Maths.safeDivision(event.count_sender_1m*1.0,event.avg_monthly_count_sender_3m_de);
  end
end
```

## Condition

```
event.count_sender_1m >= min_count
and
ratio >= min_ratio
```

## Explanation

[Transfers-EL3] - High Risk - Transaction count ( $\{event.count\_sender\_1m\}$ ) for this Sender in the last month is significantly higher ( $\{localvar.min\_ratio\} \times$ ) than Sender's monthly average count, considering behavior in the last 3 months

## Mark As

The outcome of this rule is: **high**

## Transfers-EL4

### Comment

Mismatch between sender's and receiver's BICs for transfers with amount higher than a established threshold

### Actions

- [Transfers-EL4]

### Variables

```
min_amount_th =transfers__rules_thresholds["min_amount_th"].threshold;

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency, min_amount_th);
```

### Condition

```
(event.sender_transaction_amount ?? 0 != 0) and (event.sender_bank_bic ?? '' != '') and
(event.receiver_bank_bic ?? '' != '')
and
event.sender_transaction_amount>=min_amount_th
and
event.sender_bank_bic != event.receiver_bank_bic
```

### Explanation

[Transfers-EL4] - High Risk - Transaction amount ({localvar.formatted\_amount}) exceeds the established limit ({localvar.formatted\_th}) and sender's and receiver's BICs mismatch.

### Mark As

The outcome of this rule is: **high**

## Transfers-EL6

### Comment

Sender's monthly total amount is extremely higher than average total amount (considering behavior from previous 3 months)

### Actions

- [Transfers-EL6]

### Variables

```
min_ratio =
transfers__rules_thresholds["min_ratio_sender_trx_amount_sender_monthly_avg_th"].threshold;
min_avg_amount = transfers__rules_thresholds["min_avg_monthly_amount_sender_th"].threshold;
min_count = transfers__rules_thresholds["min_count_sender_1m_th"].threshold;

ratio = match
  if (event.avg_monthly_amount_per_sender_ < min_avg_amount) then
    return -1;
  else
    return Maths.safeDivision((event.sum_amount_per_sender_1m ?? 0)*1.0,
event.avg_monthly_amount_per_sender_);
```

```
    end
end

formatted_sum = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sum_amount_per_sender_1m);
```

### Condition

```
event.count_sender_1m >= min_count
and
ratio >= min_ratio
```

### Explanation

[Transfers-EL6] - High Risk - Sum amount ({localvar.formatted\_sum}) for this sender in the last month is significantly higher ({localvar.min\_ratio} x) than Sender's monthly average amount (considering past 3 months)

### Mark As

The outcome of this rule is: **high**

## Transfers-EL9

### Comment

Mismatch between sender's and receiver's Sort Code for transfers with amount higher than a established threshold

### Actions

- [Transfers-EL9]

### Variables

```
min_amount_sort_code_th=transfers__rules_thresholds['min_amount_sort_code_th'].threshold;

formatted_amount = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sender_transaction_amount);
```

### Condition

```
(event.sender_bank_country_code ?? '' != '') and (event.receiver_bank_country_code ?? '' != '')
and
(event.sender_bank_sort_code ?? '' != '') and (event.receiver_bank_sort_code ?? '' != '')
and
(event.sender_transaction_amount ?? 0 != 0)
and
((event.sender_bank_country_code == 'GB' or event.sender_bank_country_code == 'IE')
and
(event.receiver_bank_country_code == 'GB' or event.receiver_bank_country_code == 'IE'))
and
event.sender_transaction_amount >= min_amount_sort_code_th
and
event.sender_bank_sort_code != event.receiver_bank_sort_code
```

### Explanation

[Transfers-EL9] - High Risk - Transaction amount ({localvar.formatted\_amount}) exceeds the established limit ({localvar.min\_amount\_sort\_code\_th}) and sender's and receiver's Sort Codes mismatch.

## Mark As

The outcome of this rule is: **high**

## Transfers-EL10

### Comment

Daily amount of transactions made by telephone per Sender exceeds threshold

### Actions

- [Transfers-EL10]

### Variables

```
min_sum_amount_per_sender_telephone_24h_th=transfers__rules_thresholds['min_sum_amount_per_sender_telephone_24h_th'].threshold;

formatted_sum = ValueFormatTools.formatMoney(event.sender_transaction_currency,
event.sum_amount_per_sender_telephon);

formatted_th = ValueFormatTools.formatMoney(event.sender_transaction_currency,
min_sum_amount_per_sender_telephone_24h_th);
```

### Condition

```
event.sum_amount_per_sender_telephon>= min_sum_amount_per_sender_telephone_24h_th
```

### Explanation

[Transfers-EL10] - High Risk - Transaction daily amount for this Sender ({localvar.formatted\_sum}) when channel type is telephone exceeds the established limit of {localvar.formatted\_th}.

## Mark As

The outcome of this rule is: **high**

## Transfers-EL11

### Comment

Daily transactions per sender made by telephone is extremely higher than monthly average transaction number (considering behavior from previous 3 months)

### Actions

- [Transfers-EL11]

### Variables

```
min_ratio =
transfers__rules_thresholds['min_ratio_sender_trx_count_channel_telephone_sender_monthly_avg_th']
.threshold;

min_count_trx_sender_telephone_24h_th =
transfers__rules_thresholds['min_count_trx_sender_telephone_24h_th'].threshold;

ratio = match
  if event.avg_monthly_count_sender_telep == 0.0 then
    return event.count_sender_telephone_24h;
```

```

else
    return ( event.count_sender_telephone_24h/event.avg_monthly_count_sender_telep);
end
end
end

```

### Condition

```

ratio >= min_ratio
and
event.count_sender_telephone_24h >= min_count_trx_sender_telephone_24h_th

```

### Explanation

[Transfers-EL11] - High Risk - Transaction count ({event.count\_sender\_telephone\_24h}) for this Sender when channel is telephone in last 24h is significantly higher ({localvar.min\_ratio} x) than Sender's monthly average count

### Mark As

The outcome of this rule is: **high**

## Transfers-EL12

### Comment

Daily transactions per sender made online is extremely higher than monthly average transaction number (considering behavior from previous 3 months)

### Actions

- [Transfers-EL12]

### Variables

```

min_ratio =
transfers__rules_thresholds['min_ratio_sender_trx_count_channel_online_sender_monthly_avg_3m_th']
.threshold;
min_count_trx_sender_online_24h_th =
transfers__rules_thresholds['min_count_trx_sender_online_24h_th'].threshold;

ratio = match
    if event.avg_monthly_count_sender_onlin == 0.0 then
        return event.count_sender_online_24h;
    else
        return ( event.count_sender_online_24h/event.avg_monthly_count_sender_onlin);
    end
end
end

```

### Condition

```

ratio >= min_ratio
and
event.count_sender_online_24h >= min_count_trx_sender_online_24h_th

```

### Explanation

[Transfers-EL12] - High Risk - Transaction count ({event.count\_sender\_online\_24h}) for this Sender when channel is online in last 24h is significantly higher ({localvar.min\_ratio} x) than Sender's monthly average count

**Mark As**

The outcome of this rule is: **high**